HOMEWORK 4

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7.1. Let X and Y be subspaces of a vector space V. Prove that $X \cap Y$ is a subspace of V.

Proof. Let $T = X \cap Y$. We know $T \neq \emptyset$, since both X, Y contain the zero vector, due to their vector space properties. Let $x, y \in T$. Since $x, y \in X$, by the vector space properties, $x + y \in X$. Similarly, since $x, y \in Y$, $x + y \in Y$. Therefore, vector addition is in T. Let $\alpha \in \mathbb{F}$ be a scalar. Since $x \in X$, by vector space properties, $\alpha x \in X$. Similarly, $x \in Y \implies \alpha x \in Y$. Therefore, $\alpha x \in T$. Thus, $X \cap Y$ is a vector space. □

7.2. Let V be a vector space. For $X, Y \subset V$ the sum $X + Y$ is the collection of all vectors v which can be represented as $v = x + y, x \in X, y \in Y$. Show that if X and Y are subspaces of V, then $X + Y$ is also a subspace.

Proof. Let $W = X + Y$. W is non-empty, because both X, Y contain the zero vector, due to their subspace properties ($0 + 0 = w \implies w = 0$). Let $w \in W$ such that for $x \in X, y \in Y$, $w = x + y$. Take two arbitrary vectors in W, named w_1, w_2 such that $w_1 = x_1 + y_1$ and $w_2 = x_2 + y_2$ for $x_1, x_2 \in X$ and $y_1, y_2 \in Y$. Adding both vectors: $w_1 + w_2 = x_1 + x_2 + y_1 + y_2$, where we see $x_1 + x_2 \in X$ and $y_1 + y_2 \in Y$ due to X, Y being vector spaces. Thus, vector addition in W is defined. Take an arbitrary scalar $\alpha \in \mathbb{F}$. Now let $\alpha w = \alpha x + \alpha y$. $\alpha x \in X$ and $\alpha y \in Y$ due to vector space properties. Thus, αw is defined, and W has scalar multiplication. Therefore, W is a vector space. □

7.3. Let X be a subspace of a vector space V, and let $v \in V, v \notin X$. Prove that if $x \in X$, then $x + v \notin X$.

Proof. Suppose $x + v \in X$. Let $y \in X$ such that $x + v = y$. We may rewrite this as $v = y - x$. $y - x \in X$, due to vector addition and scalar multiplication (vector space properties). But $v \notin X$. Contradiction. Therefore, $x + v \notin X$ □

7.4. Let X and Y be subspaces of a vector space V. Using the previous exercise, show that $X \cup Y$ is a subspace if and only if $X \subset Y$ or $Y \subset X$.

Proof. Suppose $X \not\subset Y$ and $Y \not\subset X$. Let $X \cup Y$ be a subspace, and let $x, y \in X \cup Y$ such that $x \in X$ but $x \notin Y$, and $y \in Y$ but $y \notin X$. Since $X \cup Y$ is a vector space, let $w \in X \cup Y$ be an arbitrary vector such that $w = x + y$. If $w \in X$, then $w - x = y$. $w - x \in X$ due to vector space properties, but $y \notin X$. Contradiction. If $w \in Y$, then $w - y = x$. $w - y \in Y$ due to vector space properties, but $x \notin Y$. Contradiction. Therefore, $X \cup Y$ is a vector space iff $X \subset Y$ or $Y \subset X$. Conversely, if $X \subset Y$, then $X \cup Y = Y$, and Y is a vector space, so $X \cup Y$ is a vector space. if $Y \subset X$, then $X \cup Y = X$, and X is a vector space, so $X \cup Y$ is a vector space. If $X \subset Y$ and $Y \subset X$, then $X = Y$, and $X \cup X = Y \cup Y$, and is automatically a

vector space. Therefore, we have shown that $X \cup Y$ is a vector space iff $X \subset Y$ or $Y \subset X$. □

2.1. Write the systems of equations below in matrix form and as vector equations:

$$\text{a)} \quad \begin{cases} x_1 + 2x_2 & -x_3 = -1 \\ 2x_1 + 2x_2 & +x_3 = 1 \\ 3x_1 + 5x_2 & -2x_3 = -1 \end{cases}$$

$$\text{b)} \quad \begin{cases} x_1 - 2x_2 & -x_3 = 1 \\ 2x_1 - 3x_2 & +x_3 = 6 \\ 3x_1 - 5x_2 & = 7 \\ x_1 & +5x_3 = 9 \end{cases}$$

Solve the systems. Write the answers in the vector form.

Proof. For the first system: $R_2 \to R_2 - 2R_1; R_3 \to R_3 - 3R_1; R_2 \to -\frac{1}{2}R_2; R_3 \to R_3 + R_2; R_3 \to -2R_3; R_1 \to R_1 + R_3; R_2 \to R_2 + \frac{3}{2}R_3; R_1 \to R_1 - 2R_2$ which gives $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{bmatrix}$. Its vector form is $4\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 3\begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

For the second system: $R_2 - = 2R_1; R_3 - = 3R_1; R_4 - = R_1; R_4 = 1/2R_4; R_3 - = R_2; R_4 - = R_2; R_1 + = 2R_2$, which gives the matrix $\begin{bmatrix} 1 & 0 & 5 & 9 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. This shows x_3 is a free variable, so let $t = x_3$. Rewriting, we get $x_1 = 9 - 5t$, $x_2 = 4 - 3t, x_3 = t$. We may write the vector form as $\begin{bmatrix} 9 \\ 4 \\ 0 \end{bmatrix} + t\begin{bmatrix} -5 \\ -3 \\ 1 \end{bmatrix} = b$ □

3.1. For what value of b the system

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ b \end{pmatrix}$$

has a solution. Find the general solution of the system for this value of b.

Proof. From RREF: $R_2 = 1/2R_2; R_2 - = R_1; R_3 - = R_1$, and since $x_3 = 1$, and $x_3 = b - 1$, then $b = 2$. Continuing on, $R_3 - = R_2; R_1 - 2R_2$. x_2 is free, let $x_2 = t$. Thus, $x_1 = -1 - 2t, x_2 = t, x_3 = 1$. The general solution is $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ □

3.2. Determine, if the vectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

are linearly independent or not.

Do these four vectors span $\mathbb{R}^4$? (In other words, is it a generating system?) What about $\mathbb{C}^4$?

Proof. Augment the vectors with the zero vector such that $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$ Doing RREF: $R_2- = R_1; R_1+ = R_2; R_3+ = R_2; R_2 = -R_2; R_4- = R_3$. Since there are free variables, this is not linearly independent. So, it cannot generate $\mathbb{R}^4$. Same with $\mathbb{C}^4$, for the same reasons. □

3.3. Determine, which of the following systems of vectors are bases in $\mathbb{R}^3$:

a) $(1, 2, -1)^T$, $(1, 0, 2)^T$, $(2, 1, 1)^T$;
b) $(-1, 3, 2)^T$, $(-3, 1, 3)^T$, $(2, 10, 2)^T$;
c) $(67, 13, -47)^T$, $(\pi, -7.84, 0)^T$, $(3, 0, 0)^T$.

Which of the systems are bases in $\mathbb{C}^3$?

Proof. From the first system, we use RREF to show linear independence due to pivots in every column and row: $R_2+ = 3R_3; R_3+ = R_1; R_2- = R_3; R_3 = 1/2R_3; R_1- = R_2; R_3- = R_2; R_1- = 2R_2$. Thus, since it is a 3×3 matrix and it is linearly independent, the vectors span $\mathbb{R}^3$ and $\mathbb{C}^3$, since $\mathbb{R} \subset \mathbb{C}$.

For the second system, $R_2+ = 3R_1; R_3+ = 2R_1; R_2 = 1/8R_2; R_3 = 1/3R_3$. Notice both R_3 and R_2 are identical. Thus, we are guaranteed a free variable. So, system 2 cannot be a basis for $\mathbb{R}^3$ nor $\mathbb{C}^3$.

Lastly, the weird system. Notice if we $R_1 \leftrightarrow R_3$, we get an lower triangle, with non-zero pivots. Thus, this system is linearly independent and spans both $\mathbb{C}^3$ and $\mathbb{R}^3$ □

3.4. Do the polynomials $x^3 + 2x$, $x^2 + x + 1$, $x^3 + 5$ generate (span) $\mathbb{P}_3$? Justify your answer.

Proof. No. This is because there are only 3 columns, where $\mathbb{P}_4$ requires 4 columns, because of the need for an $1, x, x^2$, and x^3. □

3.5. Can 5 vectors in $\mathbb{F}^4$ be linearly independent? Justify your answer.

Proof. No. Since there are 5 vectors, you will have a 4×5 matrix. Since there are more columns than rows, you will get a free variable. □

3.6. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the columns of $A^2 = AA$.

Proof. Suppose $A^2x = 0$. Let $Ax = b$, where x is some vector. Multiplying both sides, we get $A^2x = Ab = 0$. Since A is linearly independent, $Ax = 0 \implies b = 0$. Since the columns are linearly independent, $Ax = 0 \implies x = 0$. Therefore, $A^2x = 0$, showing the columns are linearly independent. □

3.7. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the rows of $A^3 = AAA$.

Proof. Using 3.6, we can apply the same technique and get $A^3x = 0$, and applying linearity, we see that $A^3x = 0 \implies x = 0$. Since A^3 is square, columns being independent $\iff$ independent rows. So the rows of A^3 are independent as well. □